Chapter 3- Frequency Analysis of Discrete Time Signals And Systems

outline

- Fourier series of discrete time periodic signal
- Power density spectrum of periodic signal
- Fourier transform of discrete time aperiodic signal
- Properties of discrete-time Fourier transform
- Sampling and quantization
- A/D and D/A converter

Classification of Fourier Pairs

	CTS-TIME	DST-TIME
PERIODIC	Continuous-Time Fourier Series (CTFS)	Discrete-Time Fourier Series (DTFS)
APERIODIC	Continuous-Time Fourier Transform (CTFT)	Discrete-Time Fourier Transform (DTFT)

Cont'd

	CTS-TIME	DST-TIME
PER	$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k F_0 t}$ $c_k = \frac{1}{T_p} \int_{T_p} x(t) e^{-j2\pi k F_0 t} dt$	$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$ $c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$
APER	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$ $X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$	$x(n) = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n} d\omega$ $X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$

The Fourier Series for Discrete-Time Periodic Signals

A given periodic sequence *x(n)* with period *N*, that is *x(n)=x(n+N)*

The Fourier representation of *x(n)* can be expressed as

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$$

where c_k ara the coefficients in the series representation. This equation is called the *Discrete-time Fourier Series* (DTFS)

The Fourier coefficients $\{c_k\}, k = 1, 2, \ldots, N-1$ provides the description of x(n) in the frequency domain

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

c_k represents the amplitude and phase associated with the frequency component

$$s_k = e^{j2\pi kn/N} = e^{j\omega_k n}$$

where $\omega_k = 2\pi k / N$

The function S_k are periodic with period N. Hence

$$s_k(n) = s_k(n+N)$$

So, that $$c_{k+N} = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi(k+N)n/N} = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} = c_k$$

Therefore, c_k is periodic sequence with fundamental period N.

Instead of focusing in periodic range *k=0,1,…, N-1* in frequency domain $0 < \omega_k = \frac{2\pi k}{N} < 2\pi$ for *0<k<N-1*, We do in range of $-\pi < \omega_k = \frac{2\pi k}{N} < \pi$, which corresponds to $-\frac{N}{2} < k < \frac{N}{2}$

Cont'd

- For discrete-time periodic signals with period N:

➢ Synthesis equation:

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$$

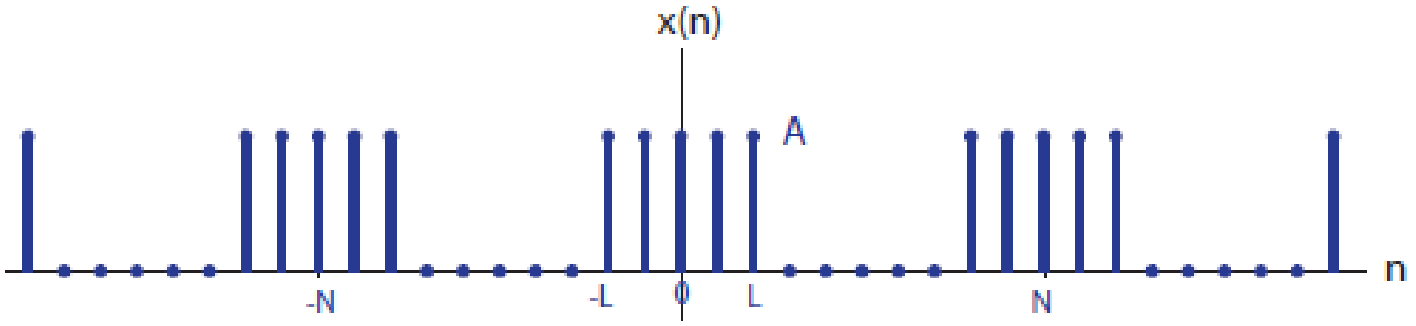
➢ Analysis equation:

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

- Ck is called the spectrum of x(n).
- Note: ck is periodic with period N.

DTFS: Example

- Find the DTFS of the following periodic square wave:



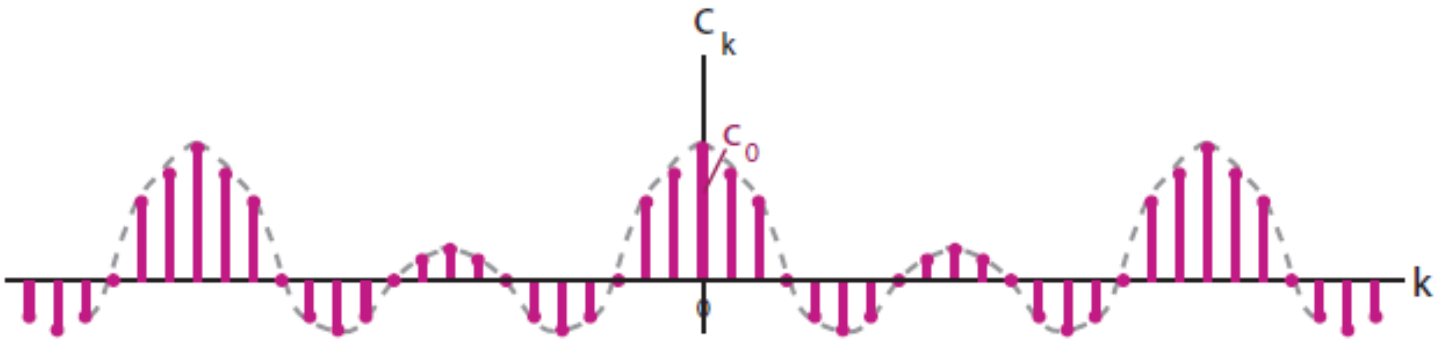
DTFS: Example(Cont'd)

$$
\begin{aligned}
c_k &= \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} = \frac{1}{N}\sum_{n=\lfloor -N/2\rfloor}^{\lfloor N/2\rfloor} x(n)e^{-j2\pi kn/N} \\
&= \frac{1}{N}\sum_{n=-L}^{L} Ae^{-j2\pi kn/N} = \frac{1}{N}\sum_{m=0}^{2L} Ae^{-j2\pi k(m-L)/N} \\
&= \frac{Ae^{j2\pi kL/N}}{N}\sum_{m=0}^{2L} e^{-j2\pi km/N} = \frac{Ae^{j2\pi kL/N}}{N}\sum_{m=0}^{2L}(e^{-j2\pi k/N})^m \\
&= \frac{Ae^{j2\pi kL/N}}{N}\frac{1-(e^{-j2\pi k/N})^{2L+1}}{1-(e^{-j2\pi k/N})} = \frac{A}{N}\frac{e^{j2\pi kL/N}-e^{-j2\pi kL/N}}{e^{j\pi k/N}-e^{-j\pi k/N}}\frac{2j}{2j} \\
&= \frac{A}{N}\frac{\sin(2\pi kL/N)}{\sin(\pi k/N)}
\end{aligned}
$$

DTFS: Example(Cont'd)

- For L = 3 and N = 18:

$$c_k = \frac{A}{18}\frac{\sin(\pi k/3)}{\sin(\pi k/18)}$$



- Note: ck is periodic with period N.

Example: Determine the spectra of the following signal

$$x(n) = \cos\frac{\pi n}{3} \qquad \omega_0 = \frac{\pi}{3} \qquad f_0 = \frac{1}{6}$$

Hence x(n) is periodic with fundamental period N=6

$$c_k = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} = \frac{1}{6}\sum_{n=0}^{5} x(n)e^{-j2\pi kn/6} \quad ,k = 0,1,...,5$$

$$x(n) = \cos\frac{\pi n}{3} = \frac{1}{2}e^{j2\pi n/6} + \frac{1}{2}e^{-j2\pi n/6}$$

$$e^{j2\pi n/6} = e^{j2\pi(5-6)n/6} = e^{j2\pi 5n/6}$$

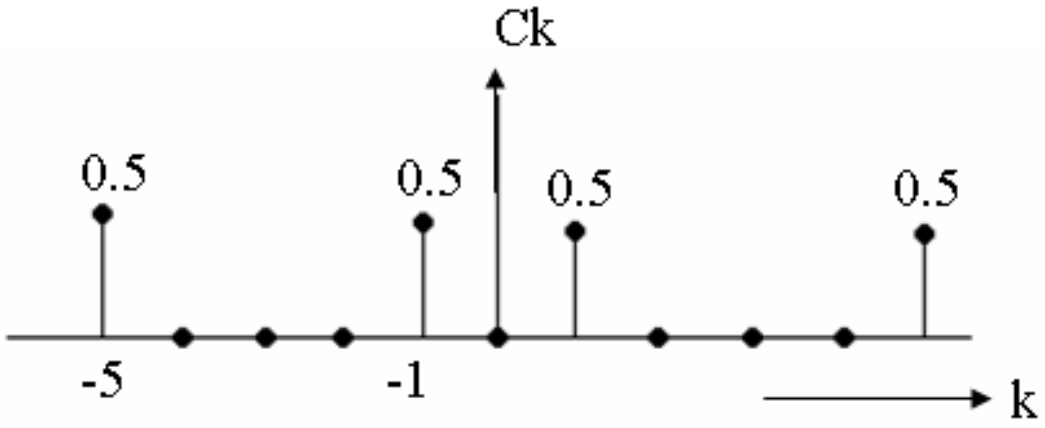
which means that $c_{-1} = c_5$

$$c_k = \frac{1}{6}\sum_{n=0}^{5} \cos(\frac{n\pi}{3})e^{-j2\pi kn/6} \quad ,k = 0,1,...,5$$

Example:(cont)

$$c_k = \frac{1}{6}\left(1 + \cos\frac{\pi}{3}e^{-j2\pi k/6} + \cos\frac{2\pi}{3}e^{-j2\pi k2/6} + \cos\frac{3\pi}{3}e^{-j2\pi k3/6} + \cos\frac{4\pi}{3}e^{-j2\pi k4/6} + \cos\frac{5\pi}{3}e^{-j2\pi k5/6}\right) \quad , \quad k = 0,1,\ldots,5$$

$c_0 = 0$ $\quad$ $c_1 = \frac{1}{2}$ $\quad$ $c_2 = 0$ $\quad$ $c_3 = 0$ $\quad$ $c_4 = 0$ $\quad$ $c_5 = \frac{1}{2}$



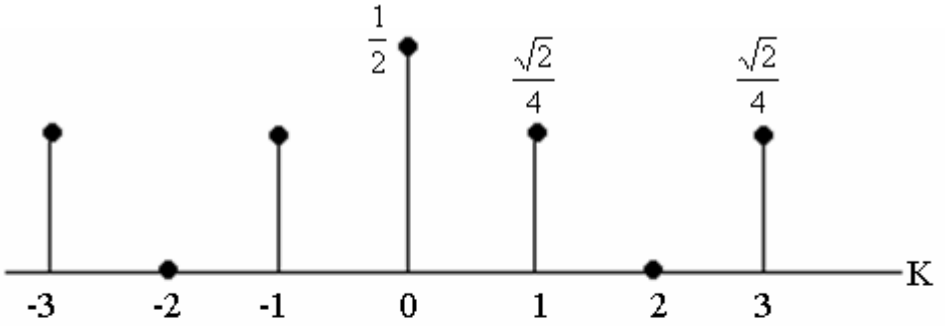
Example: Determine the spectra of the following periodic signal with period N=4

$$x(n)=\{1,1,0,0\}$$

$$c_k=\frac{1}{N}\sum_{n=0}^{N-1}x(n)e^{-j2\pi kn/N}=\frac{1}{4}\sum_{n=0}^{3}x(n)e^{-j2\pi kn/4}\quad,k=0,1,2,3$$

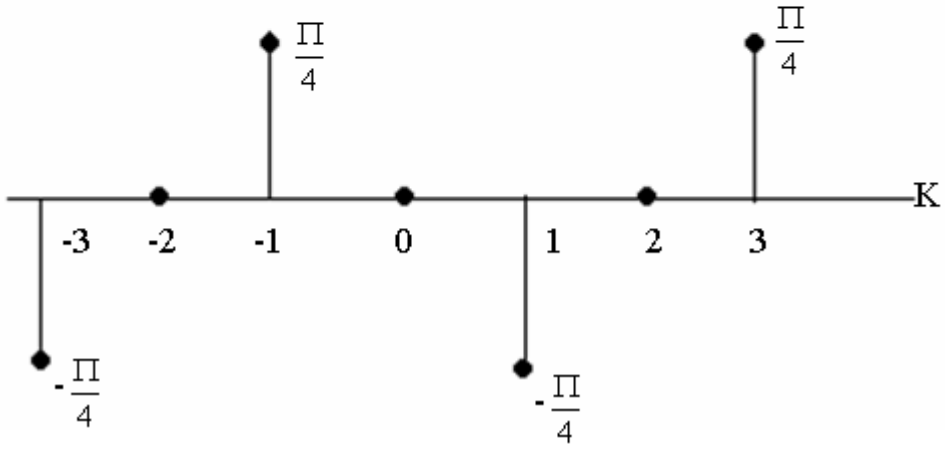
$$c_k=\frac{1}{4}(1+e^{-j2\pi k/4})\quad,\quad k=0,1,2,3$$

$$c_0=\frac{1}{2}\qquad c_1=\frac{1}{4}(1-j)\qquad c_2=0\qquad c_3=\frac{1}{4}(1+j)$$



$$c_0=\frac{1}{2}\qquad |c_1|=\frac{\sqrt{2}}{4}\qquad \angle c_1=-\frac{\pi}{4}\qquad c_2=0$$

$$|c_3|=\frac{\sqrt{2}}{4}\qquad \angle c_3=\frac{\pi}{4}$$



Example:

Determine the spectra of the signal:

$$x(n) = \text{Cos } \sqrt{2}\ \pi n$$

$$w_o = \sqrt{2}\ \pi$$

- $f_o = \frac{1}{\sqrt{2}}$ is not rational number
- Therefore, Signal is not periodic.
- Its spectra content consists of the single frequency

Power Density Spectrum of Periodic Signals

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

in terms of Fourier coefficients $\{c_k\}$

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} x(n)x^*(n) = \frac{1}{N} \sum_{n=0}^{N-1} x(n) \left(\sum_{k=0}^{N-1} c_k^* e^{-j2\pi kn/N} \right)$$

$$P_x = \sum_{k=0}^{N-1} c_k^* \left[\frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \right]$$

$$P_x = \sum_{k=0}^{N-1} |c_k|^2$$

Cont'd

- Real signals

If $x(n)$ is a real signal, $x^*(n) = x(n)$, is shown that $c_k^* = c_{-k}$ i.e. spectrum $c_k, k = 1, 2, ..., N/2$ completely describes the signal in the frequency domain.

Fourier transform of discrete time aperiodic signal (DTFT)

The Fourier transform of a finite-energy discrete time signal x(n) is defined as

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

X(ω) signal is periodic with period 2π

$$X(\omega + 2\pi k) = \sum_{n=-\infty}^{\infty} x(n)e^{-j(\omega+2\pi k)n}$$

$$X(\omega + 2\pi k) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}e^{-j2\pi kn}$$

$$X(\omega + 2\pi k) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = X(\omega)$$

The inverse Fourier transform of discrete-time signal is

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)e^{j\omega n}d\omega$$

Cont'd

- For discrete-time aperiodic signals:

➢ Synthesis equation:

$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n} d\omega$$

➢ Analysis equation:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

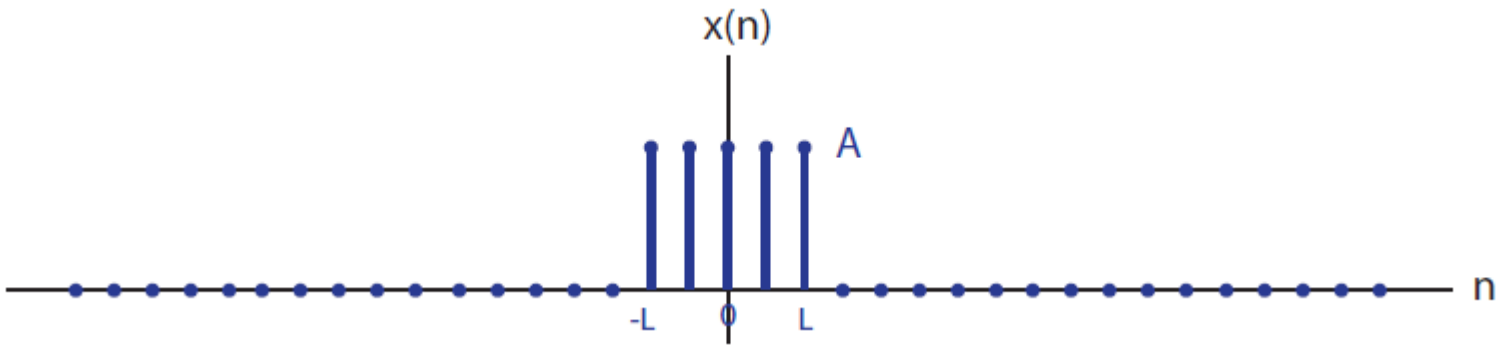
DTFT: Convergence Conditions

- DTFT exists if

$$\sum_{n=-\infty}^{\infty} |x(n)| < \infty$$

DTFT: Example

- Find the DTFT of the following rectangle function:



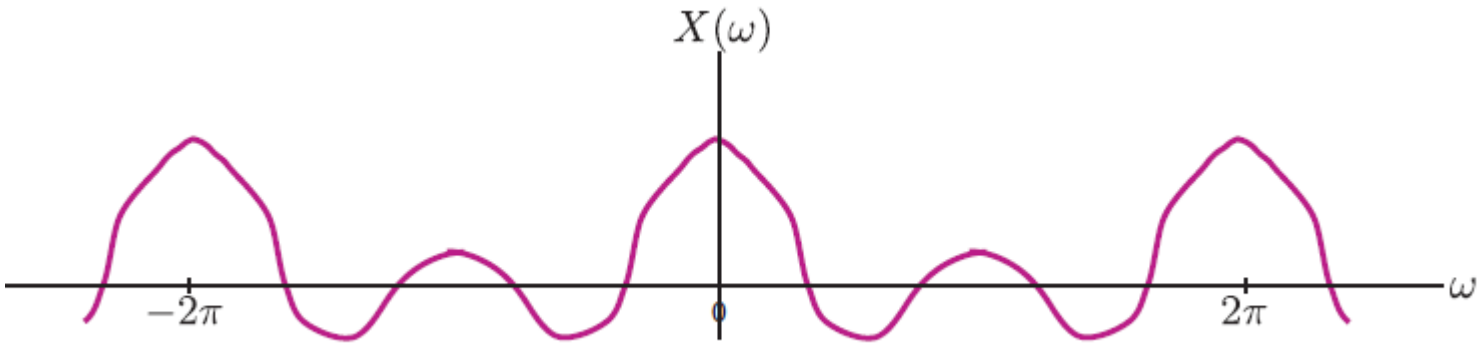
DTFT: Example(cont'd)

$$
\begin{aligned}
X(\omega) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=-L}^{L} Ae^{-j\omega n} = A\sum_{n=-L}^{L}(e^{-j\omega})^n \\
&= A\sum_{m=0}^{2L}(e^{-j\omega})^{(m-L)} = Ae^{j\omega L}\sum_{m=0}^{2L}(e^{-j\omega})^m \\
&= Ae^{j\omega L}\frac{1-e^{-j\omega(2L+1)}}{1-e^{-j\omega}}\frac{2j}{2j} \\
&= A\frac{\sin(\omega L)}{\sin(\omega/2)}
\end{aligned}
$$

DTFT: Example(cont'd)

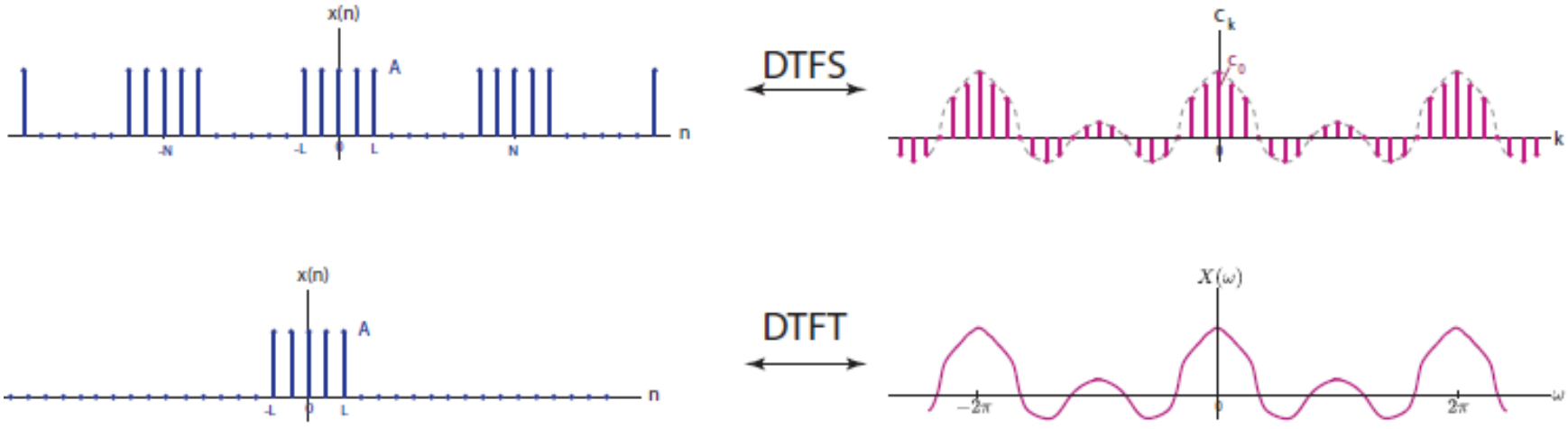
- For L = 3:

$$X(\omega) = A\frac{\sin(\omega L)}{\sin(\omega/2)}$$



- Note: X(ω) is periodic with period 2π.

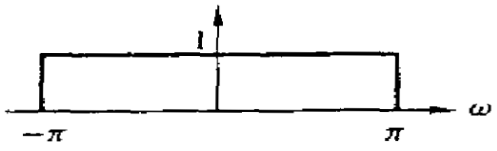
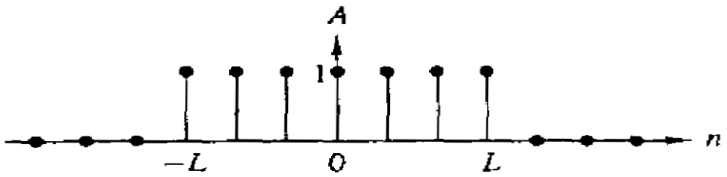
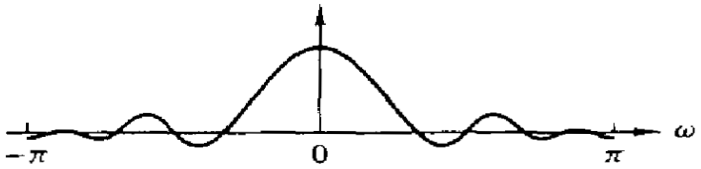
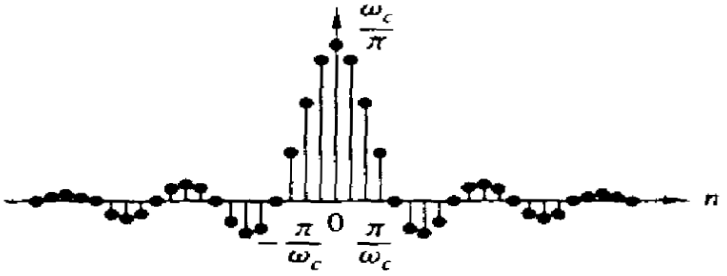
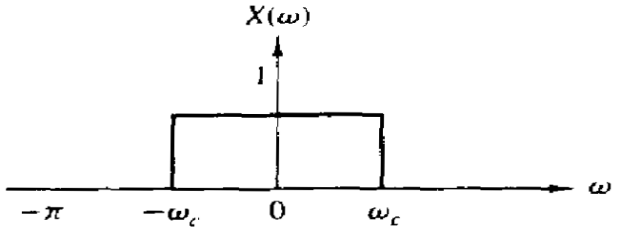
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Some Common DTFT Pairs:

Sequence	Discrete-Time Fourier Transform
$\delta(n)$	1
$\delta(n - n_0)$	$e^{-jn_0\omega}$
1	$2\pi\,\delta(\omega)$
$e^{jn\omega_0}$	$2\pi\,\delta(\omega - \omega_0)$
$a^n u(n), \quad \lvert a\rvert < 1$	$\dfrac{1}{1 - ae^{-j\omega}}$
$-a^n u(-n-1), \quad \lvert a\rvert > 1$	$\dfrac{1}{1 - ae^{-j\omega}}$
$(n+1)\,a^n u(n), \quad \lvert a\rvert < 1$	$\dfrac{1}{(1 - ae^{-j\omega})^2}$
$\cos n\omega_0$	$\pi\,\delta(\omega + \omega_0) + \pi\,\delta(\omega - \omega_0)$

Cont'd

Signal $x(n)$	Spectrum $X(\omega)$
 $x(n) = \delta(n)$	 $X(\omega) = 1$
 $x(n) = \begin{cases} A, & \lvert n\rvert \le L \\ 0, & \lvert n\rvert > L \end{cases}$	 $X(\omega) = A\,\dfrac{\sin\left(L+\frac{1}{2}\right)\omega}{\sin\frac{\omega}{2}}$
 $x(n) = \begin{cases} \frac{\omega_c}{\pi}, & n = 0 \\ \frac{\sin \omega_c n}{\pi n}, & n \neq 0 \end{cases}$	 $X(\omega) = \begin{cases} 1, & \lvert\omega\rvert < \omega_c \\ 0, & \omega_c \le \lvert\omega\rvert \le \pi \end{cases}$
$x(n) = \begin{cases} a^n, & n \ge 0 \\ 0, & n > 0 \end{cases}$	$X(\omega) = \dfrac{1}{1 - ae^{-j\omega}}$

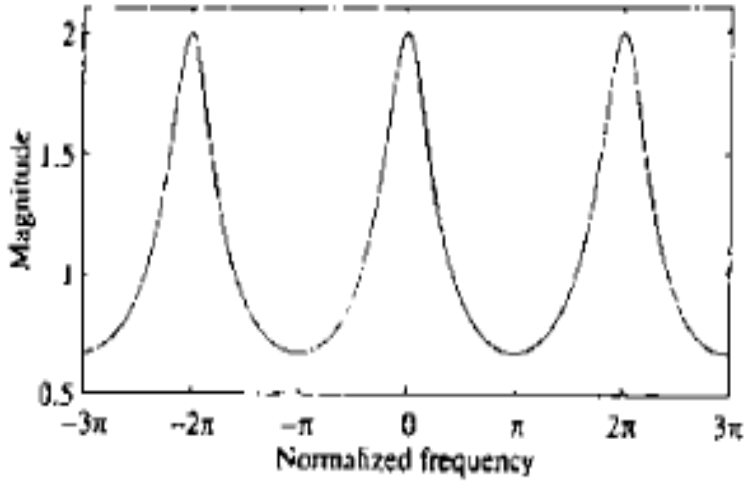
Example:

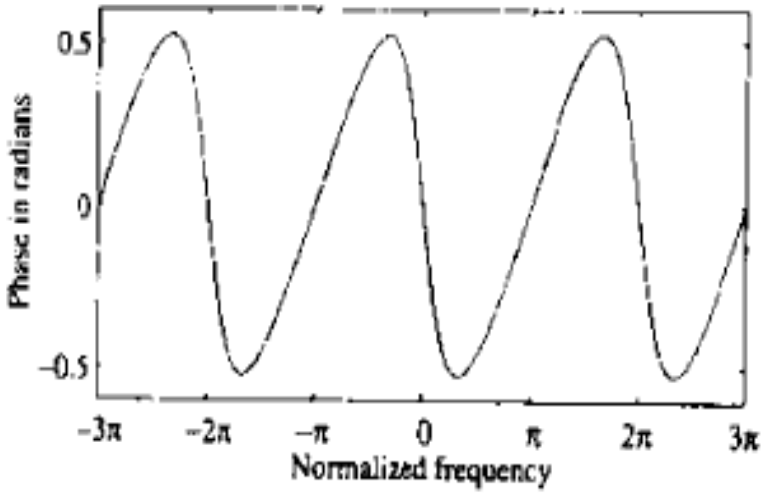
Determine a DTFT of x(n) sequence

$$x(n) = a^n u(n) \quad , \quad |a| < 1$$

$$X(w) = \sum_{n=-\infty}^{\infty} a^n u(n) e^{-jwn} = \sum_{n=0}^{\infty} a^n e^{-jwn} = \sum_{n=0}^{\infty} (ae^{-jw})^n = \frac{1}{1-ae^{-jw}}$$

For a = 0.5





DTFT Theorems and Properties

Property	Time Domain	Frequency Domain
Notation:	$x(n)$	$X(\omega)$
	$x_1(n)$	$X_1(\omega)$
	$x_2(n)$	$X_2(\omega)$
Linearity:	$a_1x_1(n) + a_2x_2(n)$	$a_1X_1(\omega) + a_2X_2(\omega)$
Time shifting:	$x(n-k)$	$e^{-j\omega k}X(\omega)$
Time reversal	$x(-n)$	$X(-\omega)$
Convolution:	$x_1(n) * x_2(n)$	$X_1(\omega)X_2(\omega)$
Correlation:	$r_{x_1x_2}(l) = x_1(l) * x_2(-l)$	$S_{x_1x_2}(\omega) = X_1(\omega)X_2(-\omega)$
		$= X_1(\omega)X_2^*(\omega)$ [if $x_2(n)$ real]
Wiener-Khintchine:	$r_{xx}(l) = x(l) * x(-l)$	$S_{xx}(\omega) = \lvert X(\omega)\rvert^2$

DTFT Symmetry Properties

Time Sequence	DTFT
$x(n)$	$X(\omega)$
$x^*(n)$	$X^*(-\omega)$
$x^*(-n)$	$X^*(\omega)$
$x(-n)$	$X(-\omega)$
$x_R(n)$	$X_e(\omega) = \frac{1}{2}[X(\omega) + X^*(-\omega)]$
$jx_I(n)$	$X_o(\omega) = \frac{1}{2}[X(\omega) - X^*(-\omega)]$
$x(n)$ real	$X(\omega) = X^*(-\omega)$ $X_R(\omega) = X_R(-\omega)$ $X_I(\omega) = -X_I(-\omega)$ $\lvert X(\omega)\rvert = \lvert X(-\omega)\rvert$ $\angle X(\omega) = -\angle X(-\omega)$
$x'_e(n) = \frac{1}{2}[x(n) + x^*(-n)]$	$X_R(\omega)$
$x'_o(n) = \frac{1}{2}[x(n) - x^*(-n)]$	$jX_I(\omega)$

Properties of discrete-time Fourier transform

Properties	Time Domain	Frequency Domain
Notation	$x(n),\ \ x_1(n),\ \ x_2(n)$	$X(\omega),\ \ X_1(\omega),\ \ X_2(\omega)$
Linearity	$ax_1(n)+b\,x_2(n)$	$aX_1(\omega)+bX_2(\omega)$
Time Shifting	$x(n-k)$	$e^{-j\omega k}X(\omega)$
Time Reversal	$x(-n)$	$X(-\omega)$
Convolution	$x_1(n)^*\,x_2(n)$	$X_1(\omega)X_2(\omega)$
Correlation	$r_{x_1x_2}(l)=x_1(l)^*\,x_2(-l)$	$S_{x_1x_2}=X_1(\omega)X_2(-\omega)=X_1(\omega)X_2^*(\omega)$
Frequency Shifting	$e^{j\omega_0 n}x(n)$	$X(\omega-\omega_0)$
Modulation	$x_1(n)\cos\omega_0 n$	$\frac{1}{2}X(\omega+\omega_0)+\frac{1}{2}X(\omega-\omega_0)$
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{2\pi}\int_{-\pi}^{\pi}X_1(\lambda)X_2(\omega-\lambda)d\lambda$
Differentiation in Frequency domain	$x^*(n)$	$j\frac{dX(\omega)}{d\omega}$
Conjugation	$nx(n)$	$X^*(-\omega)$
Parseval's Theorem	$\sum_{n=-\infty}^{\infty}x_1(n)x_2^*(n)=\frac{1}{2\pi}\int_{\pi}^{\pi}X_1(\omega)X_2^*(\omega)d\omega$	

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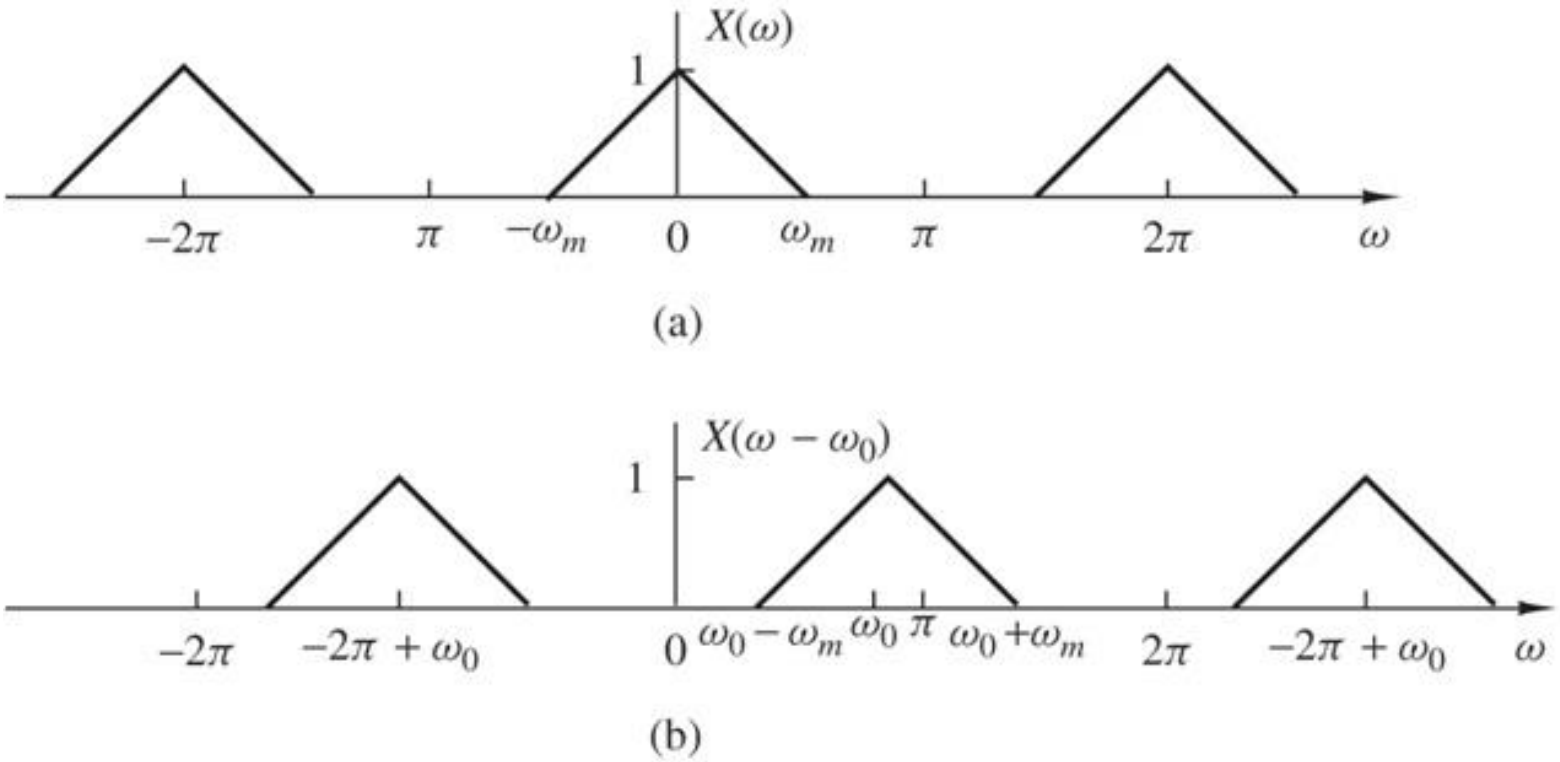


Figure Illustration of the frequency-shifting property of the Fourier transform ($\omega_0 \leq 2\pi - \omega_m$).

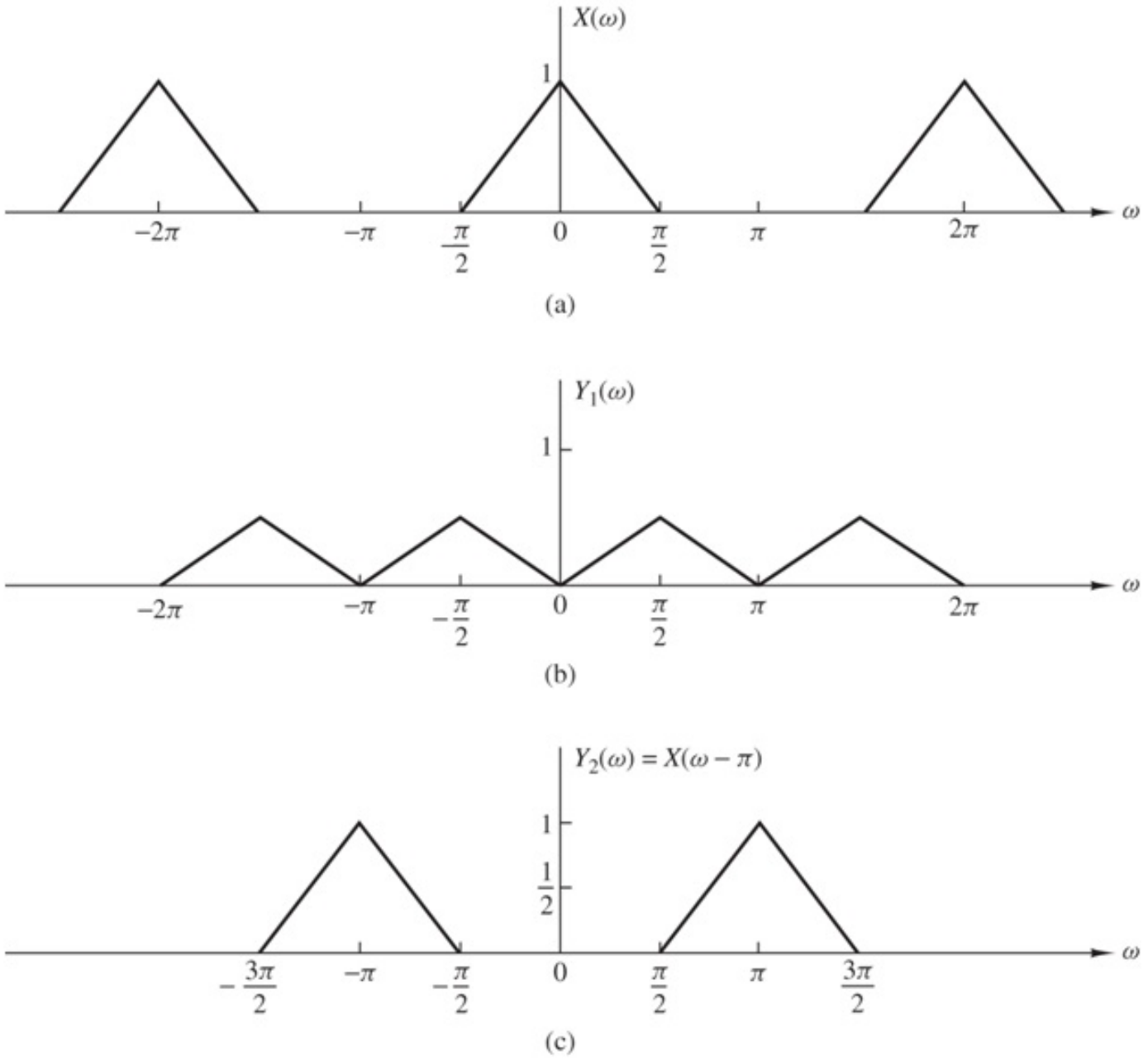


Figure Graphical representation of the modulation theorem.

Example:

Determine a DTFT of $x[n] = a^n u[n-5]$

$$x_1[n] = a^n u[n] \overset{F}{\leftrightarrow} X_1(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

$$x_2[n] = x_1[n-5] \qquad (\text{i.e.} = a^{n-5} u[n-5])$$

$$X_2(e^{j\omega}) = e^{-j5\omega} X_1(e^{j\omega}) = \frac{e^{-j5\omega}}{1 - ae^{-j\omega}}$$

$$x[n] = a^5 x_2[n] \qquad (\text{i.e.} = a^n u[n-5])$$

$$X(e^{j\omega}) = \frac{a^5 e^{-j5\omega}}{1 - ae^{-j\omega}}$$

Example:

$$x(n) = 0.8^{|n|} \qquad -\infty < n < \infty$$

$$x(n) = x_1(n) + x_2(n)$$

$$x_1(n) = \begin{cases} a^n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

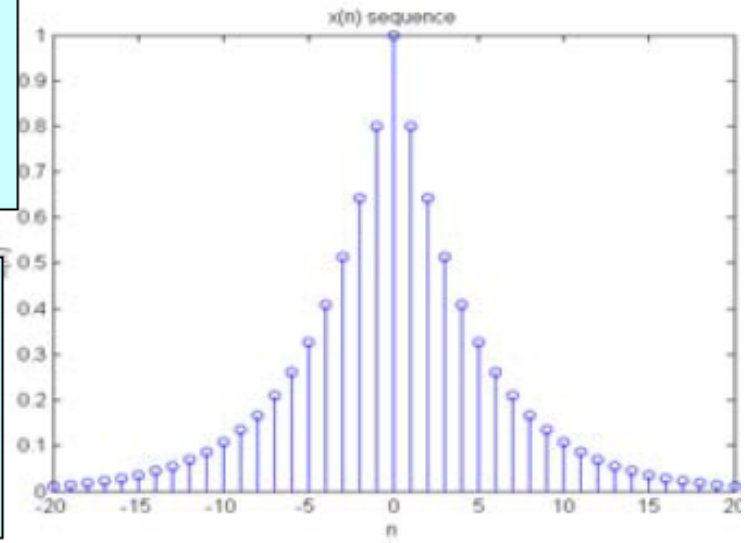
$$x_2(n) = \begin{cases} a^{-n} & n < 0 \\ 0 & n \geq 0 \end{cases}$$

$$X_1(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=0}^{\infty} 0.8^n e^{-j\omega n} = \sum_{n=0}^{\infty} (0.8e^{-j\omega})^n$$

$$= \frac{1}{1-0.8e^{-j\omega}} = \frac{e^{j\omega}}{e^{j\omega}-0.8}$$

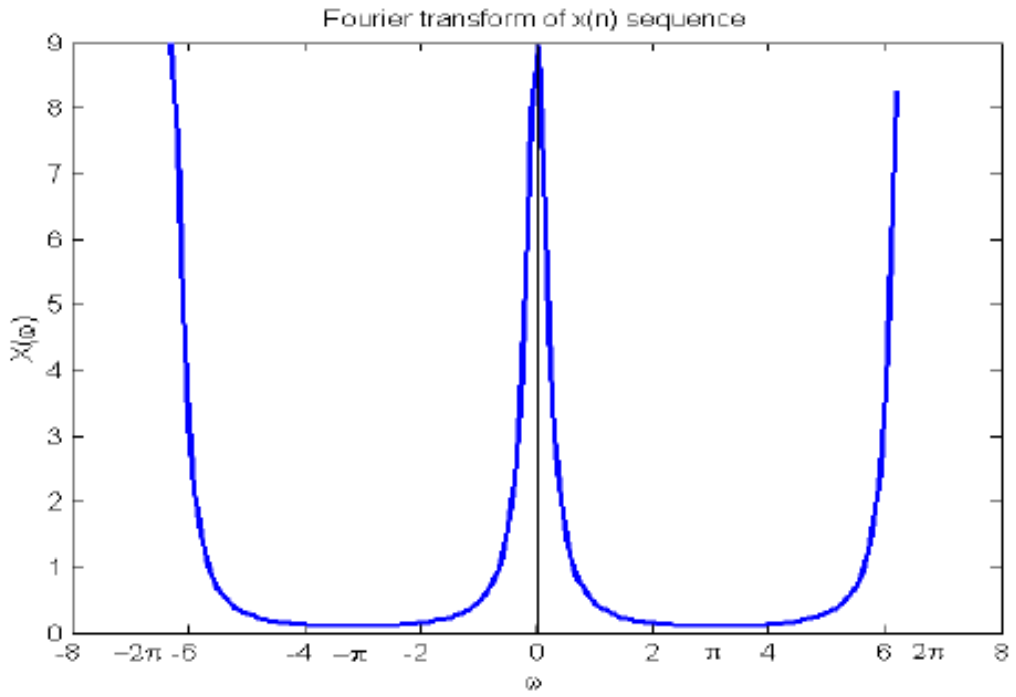
$$X_2(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=-\infty}^{-1} 0.8^{-n} e^{-j\omega n} = \sum_{n=-\infty}^{-1} (0.8e^{j\omega})^{-n}$$

$$= \sum_{n=1}^{\infty} (0.8e^{j\omega})^n = \frac{0.8e^{j\omega}}{1-0.8e^{j\omega}}$$



Example: (cont)

$$X(\omega) = X_1(\omega) + X_2(\omega) = \frac{1}{1-0.8e^{-j\omega}} + \frac{0.8e^{j\omega}}{1-0.8e^{j\omega}} = \frac{1-0.8^2}{1-2(0.8)\cos\omega + 0.8^2}$$



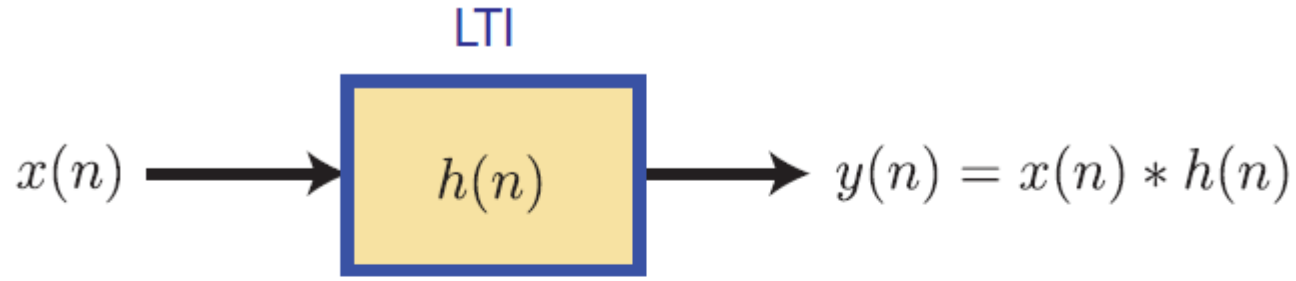

```
n=-20:20;
x=0.8.^abs(n);
stem (n,x)
xlabel ('n')
ylabel ('x(n)')
title ('x(n) sequence')

figure;
w=-2*pi:0.1:2*pi;
y=(1-0.8^2)./(1-2*0.8*cos(w)+0.8^2);
plot (w,y);
xlabel ('\omega')
ylabel ('y(\omega)')
title ('Fourier transform of x(n)
sequence')
```

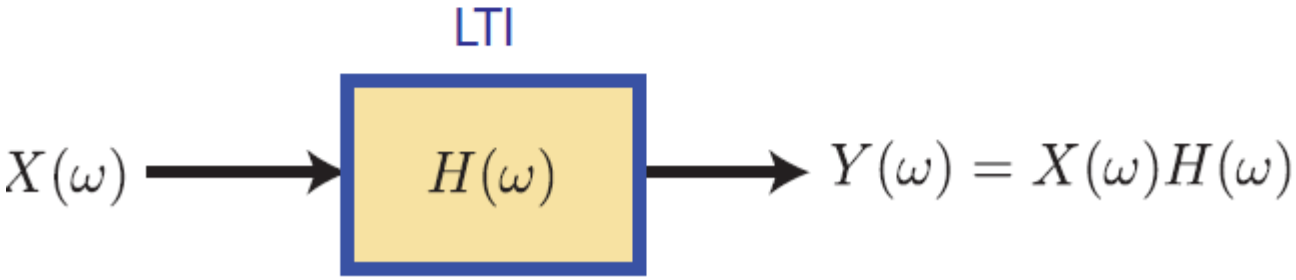
(Fourier1.m)

Frequency Domain Analysis of LTI Systems

Linear Time-Invariant (LTI) Systems



$$h(n) \longleftrightarrow H(\omega)$$



Linear Time-Invariant (LTI) Systems



- LTI: Linear Time-Invariant system
- h(n), the impulse response of an LTI systems describes the time domain c/s.
- H(ω), the frequency response describes the frequency-domain c/s.
- $h(n) \overset{F}{\leftrightarrow} H(\omega)$
- study: system response to excitation signals that are a weighted linear combination of sinusoids or complex exponentials.

- Recall the response of an LTI system to input signal x(n)

$$y(n) = \sum_{k=-\infty}^{\infty} h(n)x(n-k)$$

$$y(n) = h(n) * x(n)$$

The Frequency Response Function

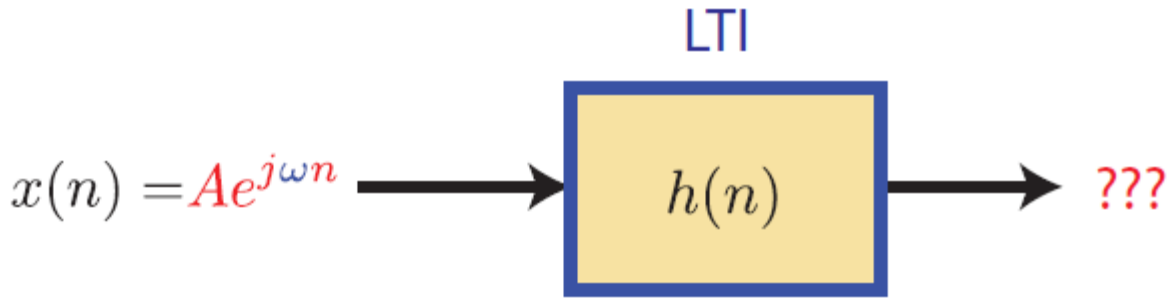
- Excite the system with a complex exponential, i.e. let $x(n) = Ae^{j\omega n}$ $-\infty < n < \infty; -\pi < \omega < \pi$

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k)$$

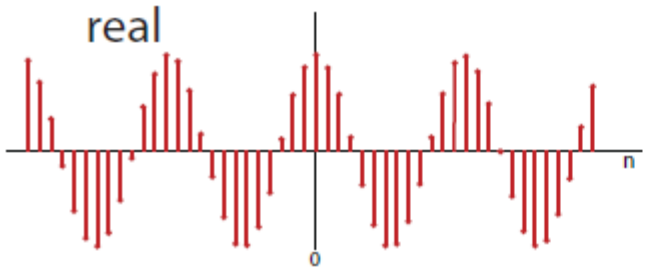
$$x(n) = Ae^{j\omega n}$$

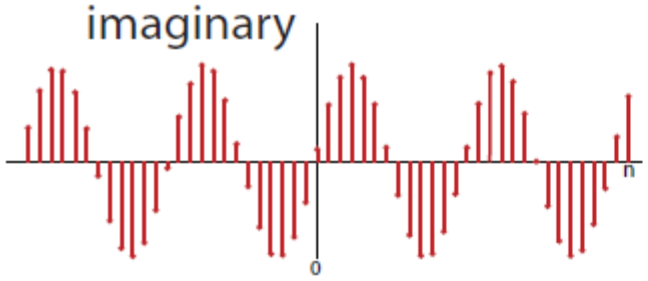
- Note: we consider x(n) to be comprised of a pure frequency of ω rad/s

Linear Time-Invariant (LTI) Systems



$$h(n) \longleftrightarrow H(\omega)$$



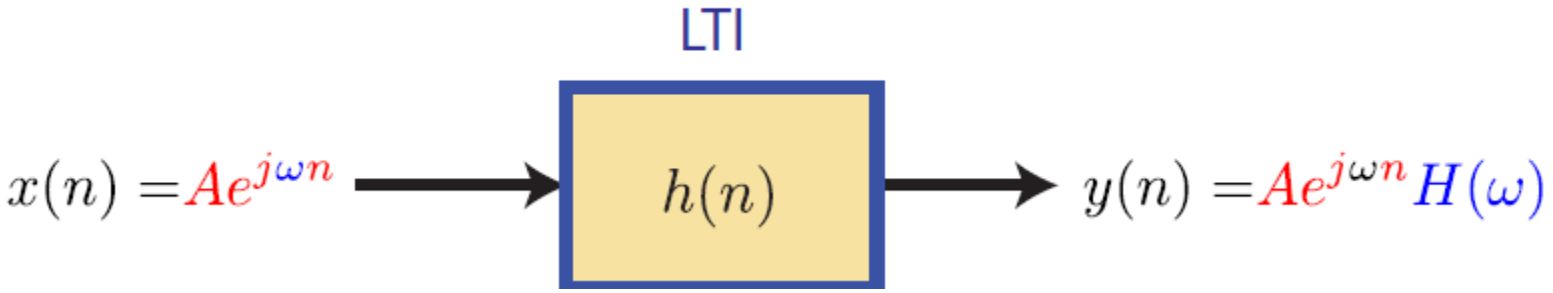


The Frequency Response Function

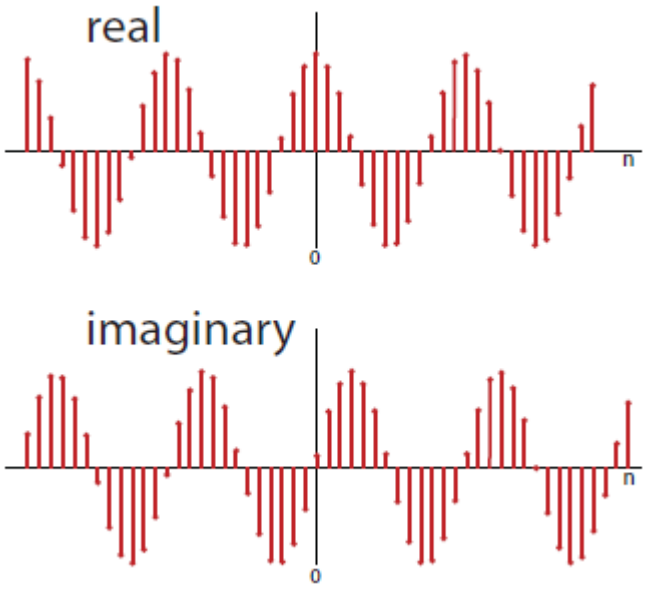
$$
\begin{aligned}
\therefore\ y(n) &= \sum_{k=-\infty}^{\infty} h(k) A e^{j\omega(n-k)} \\
&= \sum_{k=-\infty}^{\infty} h(k) A e^{j\omega n} \cdot e^{-j\omega k} \\
&= A e^{j\omega n} \cdot \underbrace{\left[\sum_{k=-\infty}^{\infty} h(k) e^{-j\omega k} \right]}_{\equiv H(\omega) = DTFT\{h(n)\}} \\
&= A e^{j\omega n} H(\omega)
\end{aligned}
$$

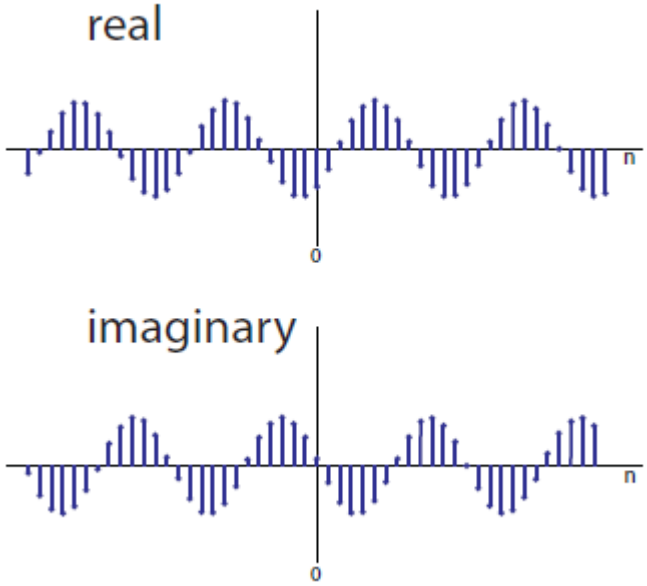
- Thus, y(n) = H(ω)x(n) when x(n) is a pure frequency.

Linear Time-Invariant (LTI) Systems



$$h(n) \longleftrightarrow H(\omega)$$





The Frequency Response Function

- Observations
  - y(n) is in the form of a complex exponential with same frequency as input, multiplied by a factor.
  - The complex exponential signal x(n) is called an eigenfunction of the system.
  - H(ω) evaluated at the frequency of the input is the corresponding eigenvalue of the system.

The Frequency Response Function

- Observations, cont.
  - As H(ω) is a Fourier transform, it is periodic with period 2π
  - H(ω) is a complex-valued function, can be expressed in polar form

$$H(\omega) = |H(\omega)|e^{\phi\omega}$$

  - h(k) is related to H(ω) through

$$h(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\omega)e^{j\omega k}\,d\omega$$

Magnitude and Phase of H(ω)

$$
\begin{aligned}
|H(\omega)| &\equiv \text{system gain for freq } \omega \\
\angle H(\omega) = \Theta(\omega) &\equiv \text{phase shift for freq } \omega
\end{aligned}
$$

$$
\begin{aligned}
y(n) &= H(\omega) A e^{j\omega n} \\
&= |H(\omega)| e^{j\Theta(\omega)} A e^{j\omega n} \\
&= A|H(\omega)| e^{j(\omega n + \Theta(\omega))}
\end{aligned}
$$

Real-valued impulse response

- An LTI system with a real-valued impulse response, exhibits symmetry properties
  - $|H(\omega)| \Rightarrow$ even function of ω.
  - $\emptyset(\omega) \Rightarrow$ odd function of ω.
  - Consequently, it is enough to know $|H(\omega)|$ and $\emptyset(\omega)$ for $-\pi < \omega < \pi$

Example

- Determine the output sequence of the system with impulse response

$$h(n) = (\frac{1}{2})^n u(n)$$

when the input signal is

$$x(n) = Ae^{j\pi n/2}, -\infty < n < \infty$$

Example, cont.

- Solution: evaluate $H(\omega)$

$$H(\omega) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

$$H(\omega = \frac{\pi}{2}) = \frac{1}{1 + j\frac{1}{2}} = \frac{2}{\sqrt{5}}e^{-j26.6^o}$$

- Thus, output is

$$y(n) = A\left(\frac{2}{\sqrt{5}}e^{-j26.6^o}\right)e^{j\pi n/2}$$

$$= \frac{2}{\sqrt{5}}Ae^{j(\pi n/2 - 26.6^o)}, -\infty < n < \infty$$

- system effect on the input: amplitude scale and phase shift
- change frequency, we change amount of scale change and phase shift.

Example: Moving Average Filter

- Determine the magnitude and phase of H(ω) for the three-point moving average (MA) system

$$y(n) = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$$

and plot these functions for $0 \leq \omega \leq \pi$

Solution:

By inspection,

$$h(n) = \frac{1}{3}\delta(n+1) + \frac{1}{3}\delta(n) + \frac{1}{3}\delta(n-1)$$

Example: Moving Average Filter (Cont.)

$$h(n) = \{\frac{1}{3}, \underline{\frac{1}{3}}, \frac{1}{3}\}$$

consequently,

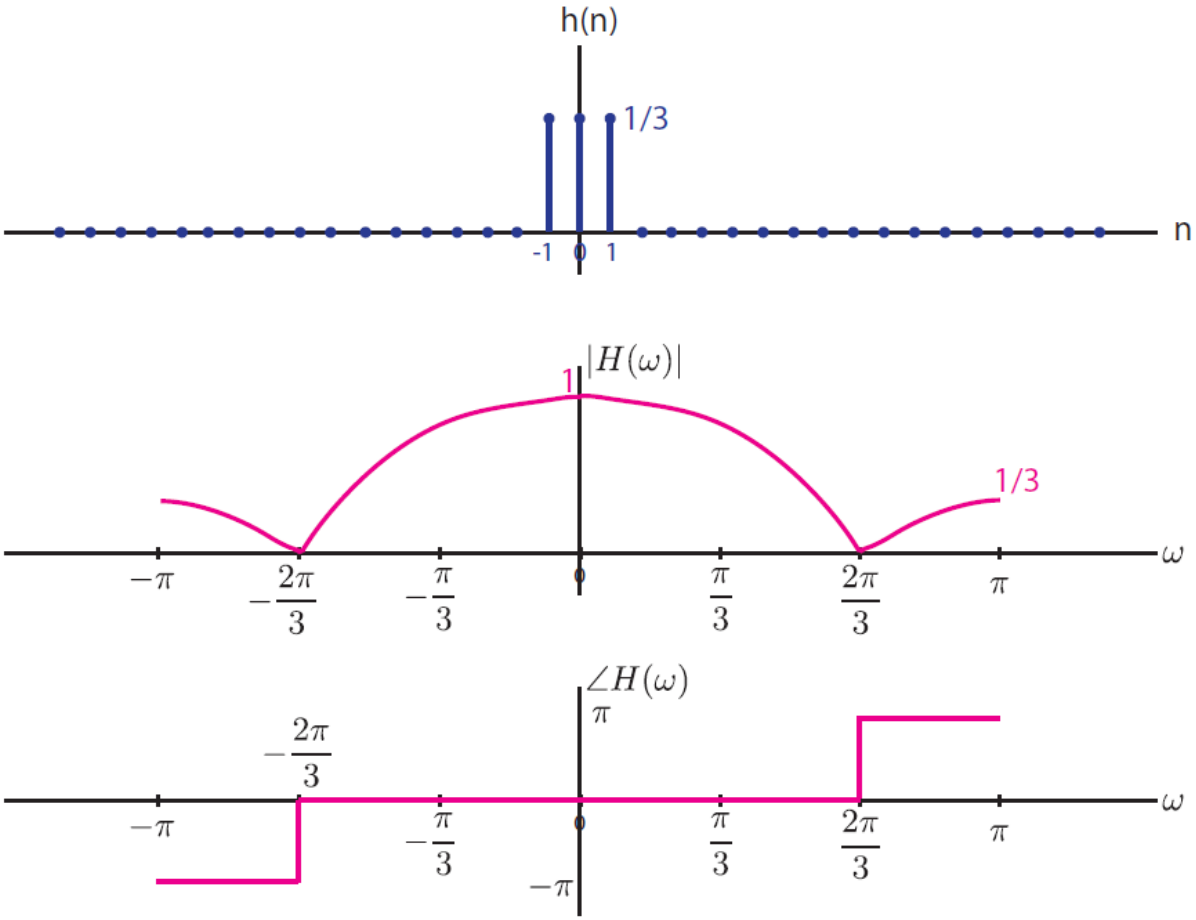
$$H(\omega) = \sum_n h(k)e^{-j\omega k} = \frac{1}{3}(e^{j\omega} + 1 + e^{-j\omega}) = \frac{1}{3}(1 + 2\cos\omega)$$

Hence

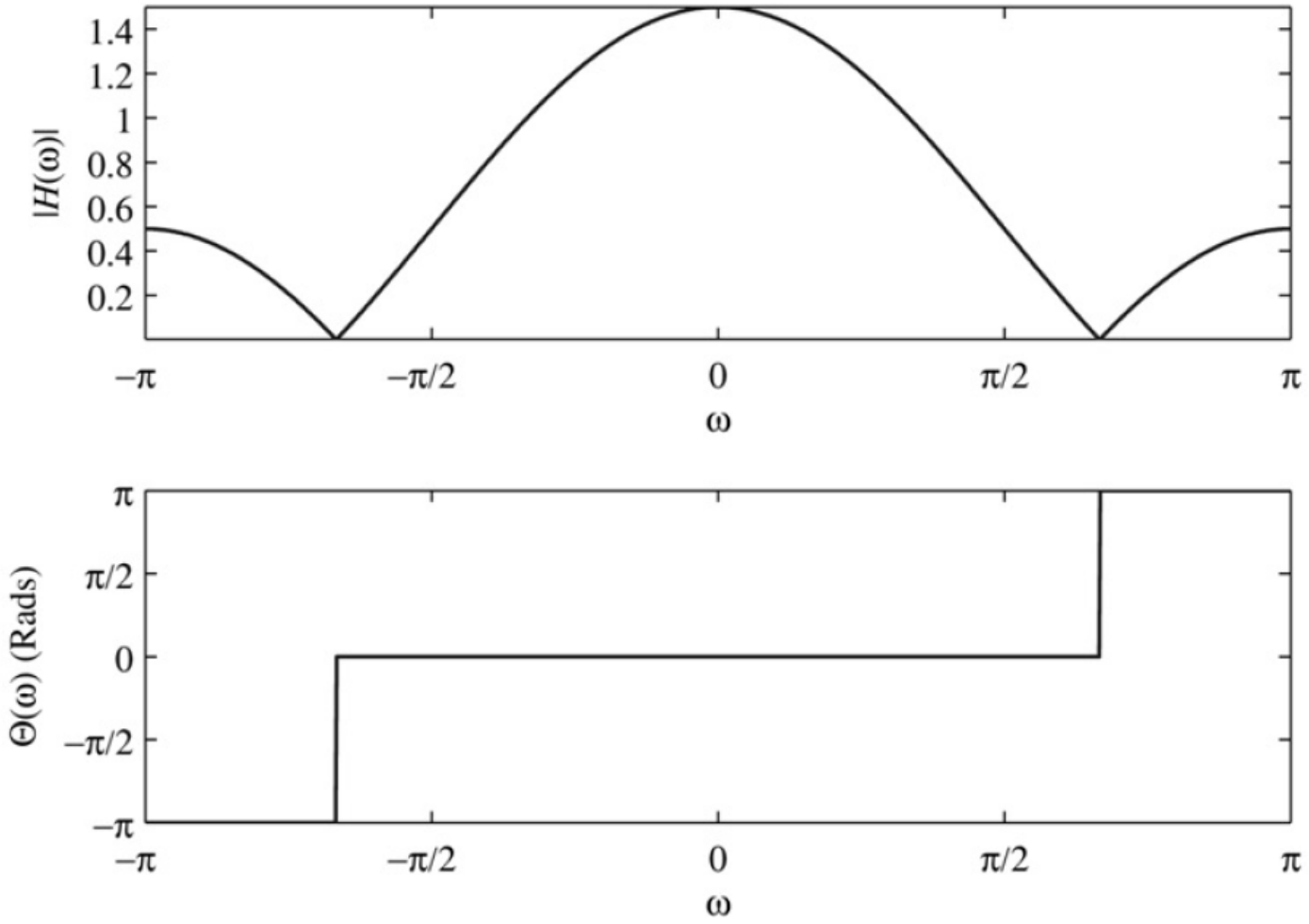
$$|H(\omega)| = \frac{1}{3}|1 + \cos\omega|$$

$$\phi(\omega) = \begin{cases} 0, 0 \leq \omega \leq \dfrac{2\pi}{3} \\ \pi, \dfrac{2\pi}{3} \leq \omega < \pi \end{cases}$$

Example: Moving Average Filter (Cont.)



Example: Moving Average Filter (Cont.)



Magnitude and phase responses for the MA system in Example

Example: system response to sinusoids

- Determine the response of the system with impulse response:

$$h(n) = (\frac{1}{2})^n u(n)$$

when the input signal is

$$x(n) = 10 - 5\sin\frac{\pi}{2}n + 20\cos\pi n, -\infty < n < \infty$$

Recall, the frequency response of the system is

$$H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

Example, cont.

- Solution
- Idea Recognise the frequency of each part of the input signal, and find corresponding system response.
- First term 10, fixed signal $\Rightarrow \omega = 0$

$$H(0) = \frac{1}{1 - \frac{1}{2}} = 2$$

- $5 \sin \frac{\pi}{2} n$ has a frequency $\omega = \pi/2$, thus

$$H(\frac{\pi}{2}) = \frac{2}{\sqrt{5}} e^{-j26.6^o}$$

- $20 \cos \pi n$ has a frequency $\omega = \pi$

$$H(\pi) = \frac{2}{3}$$

Example, cont.

- Hence, the response of the system to $x(n)$ is

$$y(n) = 20 - \frac{10}{\sqrt{5}} \sin\left(\frac{\pi}{2}n - 26.6^{\circ}\right) + \frac{40}{3}\cos \pi n \qquad -\infty < n < \infty$$

The General Case

- Most general case: input to the system is an arbitrary linear combination of sinusoids of the form

$$x(n) = \sum_{i=1}^{L} A_i \cos(\omega_i n + \phi_i), -\infty < n < \infty$$

Where $\{A_i\}$ and $\{\emptyset_i\}$ amplitude and phase of corresponding sinusoidal component i .

- System response will be of the form:

$$y(n) = \sum_{i=1}^{L} A_i |H(\omega_i)| \cos\left[\omega_i n + \phi_i + \Theta(\omega_i)\right]$$

Where $|H(\omega_i)|$ and $|\Theta(\omega_i)|$ are the magnitude and phase imparted by the system to the individual frequency components of the input signal.

Example:

- A linear time-invariant system is described by the following difference equation

$$y(n) = ay(n-1) + bx(n) \qquad 0 < a < 1$$

a) Determine the magnitude and phase of the frequency response H(ω) of the system.

- The impulse response of the system is

$$h(n) = ba^n u(n)$$

Example: cont.

- The Frequency response is $H(\omega) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$

$$= \frac{b}{1 - ae^{-j\omega}}$$

$$|1 - ae^{-j\omega}| = \sqrt{(1 - a\cos\omega)^2 + (a\sin\omega)^2}$$

$$= \sqrt{1 + a^2 - 2a\cos\omega}$$

$$\measuredangle(1 - ae^{-j\omega}) = \tan^{-1}\frac{a\sin\omega}{1 - a\cos\omega}$$

$$|H(\omega)| = \frac{|b|}{\sqrt{1 + a^2 - 2a\cos\omega}}$$

$$\measuredangle H(\omega) = \Theta(\omega) = \measuredangle b - \tan^{-1}\frac{a\sin\omega}{1 - a\cos\omega}$$

Example: cont.

b) Determine the output of the system to the input signal

$$x(n) = 5 + 12 \sin \frac{\pi}{2} n - 20 \cos \left(\pi n + \frac{\pi}{4} \right)$$

The input signal consists of components of frequencies $\omega = 0, \pi/2$, and π. For $\omega = 0$, $|H(0)| = 1$ and $\Theta(0) = 0$. For $\omega = \pi/2$,

$$\left| H\left(\frac{\pi}{2}\right) \right| = \frac{1-a}{\sqrt{1+a^2}} = \frac{0.1}{\sqrt{1.81}} = 0.074$$

$$\Theta\left(\frac{\pi}{2}\right) = -\tan^{-1} a = -42^\circ$$

For $\omega = \pi$,

$$|H(\pi)| = \frac{1-a}{1+a} = \frac{0.1}{1.9} = 0.053$$

$$\Theta(\pi) = 0$$

Example: cont.

Therefore, the output of the system is

$$
\begin{aligned}
y(n) &= 5|H(0)| + 12\left|H\left(\frac{\pi}{2}\right)\right| \sin\left[\frac{\pi}{2}n + \Theta\left(\frac{\pi}{2}\right)\right] \\
&\quad - 20|H(\pi)| \cos\left[\pi n + \frac{\pi}{4} + \Theta(\pi)\right] \\
&= 5 + 0.888 \sin\left(\frac{\pi}{2}n - 42^\circ\right) - 1.06 \cos\left(\pi n + \frac{\pi}{4}\right) \qquad -\infty < n < \infty
\end{aligned}
$$